

Sean Sheng-Tse Ru

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EDUCATION

Georgia Institute of Technology - Atlanta, GA

M.S. in Computer Science (Visual Analytics)

May 2025

B.S. in Computer Science (Database/Networking Technologies and Artificial Intelligence)

May 2024

EXPERIENCE

Data Analyst @ Capital One

Aug. 2025 - Present

Data Analyst Intern @ Capital One

June 2024 - Aug. 2024

- Developed a Tableau dashboard of metrics, graphs, and tables, enabling internal clients to **filter and analyze over 2 years of Bankruptcy Complaints data**
- Collaborated with compliance testers, risk management, and business analysts to define output columns and parameters to **design and develop SQL and Python scripts** for pulling and filtering Complaints and COAF Account Origination data
- Partnered with software engineers on UI enhancements, documenting dependencies and suggesting improvements to **streamline the onboarding process of test scripts** to the internal testing platform for future Data Analysts

Researcher @ The Georgia Tech Visualization Lab

Feb. 2024 - Present

beautiVis Dataset:

- Built a scalable, automated pipeline to source and structure over 50k community-curated visualizations** from r/dataisbeautiful subreddit using BeautifulSoup for image extraction and post-processing scripts to standardize metadata
- Engineer advanced prompts for ChatGPT-4 to **automate the extraction of visualization design data**, resulting in a comprehensive dataset of visualizations and their properties

Charting and Quantifying Visual Complexity:

- Developed Visualization Taxonomy/Rubric for labeling the MASSVIS dataset and manually labeled 2000+ visualizations
- Drafted papers on labeling considerations for quantifying visual complexity paper submitted to 2024 IEEE VIS
- Submitted an accepted 2024 IEEE VIS poster** about Quantifying Visual Complexity with R statistical analysis on the variation of chart types' visual complexity ratings

Graduate Teaching Assistant @ College of Computing

Aug. 2024 - Present

CS 4460 Intro to Information Visualization - Spring 2025:

- Supported 100+ students** through office hours and by grading quizzes, exams, and homework

CS 3001 Computing & Society - Fall 2024

- Led engaging discussions with two groups of 10–15 students on **ethical and societal implications of technology** and graded papers and exams, delivering timely and constructive feedback

Head Teaching Assistant @ School of Industrial and Systems Engineering

Aug. 2023 - Present

- Supported 400+ students** in understanding course material, focusing on probability, probability distributions, point estimation, confidence intervals, hypothesis testing, and linear regression
- Held office hours, responded to emails, supported students' understanding of course material and use of statistical software (e.g., R), and achieved a **90% effectiveness rating** in the Course Instructor Opinion Survey
- Evaluated and graded exams, homework, and class participation, and created comprehensive answer keys for exams to ensure fair and consistent assessment

Data Science Intern @ Optimized Payments

May 2023 - Aug. 2023

- Took Merchant Advisory Group Course 101 to understand the Financial Transaction and Payment Industry
- Wrote Python Script to extract data from PDF Files using Tabula
- Used Tableau to **visualize and analyze interchange fees and sales data** (Data Visualization) for 4 clients
- Performed data mapping in Snowflake and DBeaver, and developed a Python Script to compare data between SQL and Snowflake to find discrepancies in data immigration to Snowflake

SKILLS

Python (Pandas, NumPy, Seaborn, TensorFlow, Plotly, PyTorch, scikit-learn, Tabula), **SQL** (PostgreSQL, MySQL, DBeaver), **R** (ggplot2, quantmod, Janitor, Skimr, Tidyverse, DT), **HTML/CSS**, **JavaScript** (D3.js), **C/C++**, **Java** (JavaFx, Android Studies), **Tableau**, **Snowflake**, **CI/CD** (Git)